

PAPER_1762_SF_EFFICIENCY_1_75

Daniel T. Murphy

PAPER_1762 — Star Formation Efficiency Boost = SF efficiency = $K_{\text{MEX}} \times \Phi_{5/6} = 1.736$ (resolves cooling-flow puzzle)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Astrophysics

Date: June 18, 2026

Status: CLOSED — 0.79% closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1334 (PAPER_132x-135x corpus) gives:

``

SF efficiency = $K_{\text{MEX}} \times \Phi_{5/6} = 1.736$ (resolves cooling-flow puzzle)

``

Status: **0.79%**.

UQFF Closed Identity

``

SF efficiency = $K_{\text{MEX}} \times \Phi_{5/6} = 1.736$ (resolves cooling-flow puzzle) 0.79%

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1334

- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)

- Calculator dispatch: `calculate_paradox({"paradox": "sf_efficiency_1_75"})`

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